

Use Case 11 - Compliance Audit and Hierarchical Forecasting Research

NPN_HACKATHON - Walmart M5 store-item demand forecasting

Stage 7, Experiments #80-#84

Verdict: KEEP CHAMPION (0.60 direct + 0.40 recursive, RMSE 2.0929)

Project: NPN_HACKATHON — Walmart M5 store-item demand forecasting **Branch:** Stage 7, Experiments #80–#84 **Scope:** audit the shipped system against every clause of Use Case 11, then investigate the one requirement it did not yet satisfy. The protected champion was never modified, retrained over, or replaced.

Use Case 11. *“Build a forecasting model that handles hierarchical aggregation, external covariates (price/promo/holiday) and intermittent demand, producing accurate 28-day-ahead forecasts per store/item.”*

1. Verdict

B. KEEP CHAMPION.

The shipped model stands unchanged:

```
0.60 x Direct LightGBM Tweedie(1.1), 38 features
0.40 x Recursive LightGBM Tweedie(1.1), 32 features
primary window RMSE 2.0929 MAE 1.0395 WAPE 0.7205 bias -0.0224
```

Three new directions were investigated. Two were rejected before any model was trained, on measured headroom. The third — formal hierarchical reconciliation — was trained and validated across four disjoint windows under **two** independent protocols, and both were rejected on the same pre-registered mechanism criterion, despite each improving mean RMSE, mean MAE and mean high-volume RMSE simultaneously.

Experiment	Protocol	Wins	Mean ΔRMSE	Mechanism	Decision
#81 FULL	α fixed at inner-window optimum	3/4	−0.0074	2/4	REJECTED
#81 Demeaned	α fixed at inner-window optimum	3/4	−0.0088	2/4	REJECTED
#82 FULL	α per origin, preceding window	3/4	−0.0061	2/4	REJECTED
#82 Demeaned	α per origin, preceding window	3/4	−0.0093	2/4	REJECTED

Section 6 explains why that rejection is the right call, and Section 6.1 shows that the natural fix — choosing α adaptively from data available at forecast time — moves α the **wrong way** on the one window where it matters.

The champion was reproduced from scratch on all four evaluation windows during this branch and matched its recorded scores to four decimals every time (2.0929 / 2.1547 / 2.1074 / 2.1496). That is an independent reproducibility result, not a claim inherited from earlier stages.

2. Requirement gap analysis

Assessed against the code, features, artefacts and experiment records as they stood **before** this branch began.

#	Requirement	Status before	Why
A	Hierarchical aggregation	~ PARTIAL	The hierarchy entered the model only as five categorical features (item_id, dept_id, cat_id, store_id, state_id). No forecast was ever produced at an aggregate level, no coherence property was stated, no reconciliation was attempted, and aggregate accuracy was quoted at four ad-hoc levels in the README without a per-level measurement.
B1	Price covariates	✓ SATISFIED	sell_price (target-day), recent_avg_price (origin-relative), price_rel_to_recent_avg, price_is_missing. Price dynamics were additionally tested and rejected (opt_02_v2_C_price, 2.1281 vs 2.1210).
B2	Promotion covariates	~ PARTIAL	M5 contains no promotion field. Nothing in calendar.csv, sales_train*.csv or sell_prices.csv flags a promotion, display, feature ad or coupon. Promotion is observable only through price.
B3	Holiday / calendar covariates	✓ SATISFIED	event_name_1/2, event_type_1/2, wday, month, year, is_weekend, plus state-matched SNAP.
C	Intermittent demand	~ PARTIAL	Tweedie(1.1) selected by a power sweep, and the hurdle family tested twice and rejected. But Croston, SBA and TSB — named in the brief — had never been tested, and adequacy had only been checked in aggregate, never regime by regime.
D	28-day-ahead forecasting	✓ SATISFIED	Fixed-origin direct 28-day design; every history feature frozen at the origin; 46 foundation checks; corruption test overwriting all post-origin sales with 9999 and requiring bit-identical features.
E	Per store-item forecasting	✓ SATISFIED	30,490 series × 28 days = 853,720 predictions, none excluded, none weighted.

Three gaps therefore drove this branch: **A** (no hierarchical forecasting at all), **B2** (promotion never represented explicitly) and **C** (classical intermittent methods never measured).

3. Hierarchical aggregation

3.1 Coherence

The system forecasts only the bottom level, so every aggregate is a sum of bottom-level forecasts. That is bottom-up reconciliation and it is **trivially coherent** — there is no incoherence to repair, because no independent aggregate forecast exists to disagree with. Coherence was never the open question.

The open question was whether information living at an aggregate level could reduce **bottom-level** error, which is what the project is scored on.

3.2 Aggregate accuracy at all 12 M5 levels

Measured from the champion's own residuals (uc11_hierarchy_levels.csv):

Level	Groups	Bottom-up WAPE	Accuracy	Common share of squared error
L1 total	1	0.0283	97.2%	0.06%
L2 state	3	0.0396	96.0%	0.13%
L3 store	10	0.0530	94.7%	0.23%
L4 category	3	0.0379	96.2%	0.12%
L5 department	7	0.0518	94.8%	0.26%
L6 state × cat	9	0.0514	94.9%	0.23%
L7 state × dept	21	0.0680	93.2%	0.44%
L8 store × cat	30	0.0692	93.1%	0.42%
L9 store × dept	70	0.0891	91.1%	0.78%
L10 item	3,049	0.2920	70.8%	16.76%
L11 item × state	9,147	0.4446	55.5%	36.36%
L12 store-item	30,490	0.7236	27.6%	—

"Common share" is exact, not estimated. For a group of n members with discrepancy D , the bottom-level squared error splits as $\sum(e^2) = D^2/n + \sum((e - D/n)^2)$. The first term is the entire portion any aggregate-level information could ever address; the second is idiosyncratic and provably invisible to it.

3.3 Oracle bounds — why the coarse levels were rejected without training

Each oracle is handed the **true** aggregate and allowed to redistribute it perfectly. No reconciliation method can beat its own oracle.

Level	Equal-share	Proportional (top-down)	In-sample-fitted shares
L1 total	−0.0007	−0.0024	−0.0655
L3 store	−0.0024	−0.0100	−0.0985
L5 department	−0.0028	−0.0047	−0.1060
L9 store × dept	−0.0084	−0.0221	−0.1412
L10 item	−0.1864	−0.2272	−0.4004
L11 item × state	−0.4331	−0.5154	−0.6868

Knowing the true chain total **perfectly** is worth −0.0007 RMSE. Knowing the true store × department daily total **perfectly** is worth −0.0221. **Every level above item is dead**, and no amount of modelling effort changes that, because the information

simply is not there. That result alone retired bottom-up, top-down, middle-out and MinT at those levels without a single training run.

The item-bearing levels are different, and were investigated properly.

3.4 Experiments #80 / #80b / #80c — selection and controls (inner window)

All selection was done on the inner window (origin d_1885, targets d_1886–d_1913), strictly before the primary window's targets and before the final forecast origin.

- **Objective.** Tweedie(1.1), which wins at the bottom level, is clearly **wrong** at the aggregate level: item-level RMSE 10.79 versus 8.18 for L2. Aggregates are neither sparse nor zero-inflated. L2 was selected.
- **Level.** item (−0.0160) beat item × state (−0.0137) and store × dept (−0.0122).
- **Leakage.** A corruption test rebuilt all 30 aggregate features from a panel whose post-origin sales were replaced with 9999. All 30 came back identical; the target changed, so the test could not pass vacuously.
- **Negative controls.** Three were run. One fired: an oracle global rescale was worth −0.0230, **more** than the method's −0.0160.

That last result forced Experiment #80c, which decomposed the correction:

Variant	RMSE	ΔRMSE
champion	2.0677	—
global rescale (oracle)	2.0447	−0.0230
full reconciliation	2.0517	−0.0160
item-specific component only	2.0624	−0.0053
rescale + item-specific	2.0415	−0.0262

Two thirds of the headline gain was the item model happening to undo a level anomaly specific to that window: the champion over-forecasts the inner window by **+0.0817 units per row**, while on the primary window it is calibrated (bias −0.0224, and the same rescale oracle is worth −0.0000 there). Promoting on the headline number would have shipped a calibration artefact fitted to one atypical month.

Both variants — FULL and Demeaned (global component divided out, so it can only redistribute between items) — were carried forward with α fixed at their inner-window optima, 0.55 and 0.35.

3.5 Experiment #81 — four-window validation

$$P'(s,d) = \text{clip}(P(s,d) + a * (P(s,d) / F(i,d)) * (\hat{A}(i,d) - F(i,d)), 0, \text{None})$$

which is forecast-proportions top-down of the item level blended into the bottom-up forecast — textbook middle-out reconciliation.

FULL, $\alpha = 0.55$

Window	Champion	Reconciled	ΔRMSE	ΔMAE	ΔHigh-vol
primary_spring_2016	2.0929	2.0913	−0.0017	−0.0074	+0.0028
christmas_2015	2.1547	2.1705	+0.0158	+0.0001	+0.0820
summer_2015	2.1074	2.0888	−0.0187	−0.0156	−0.0728

Window	Champion	Reconciled	Δ RMSE	Δ MAE	Δ High-vol
autumn_2015	2.1496	2.1246	-0.0250	-0.0127	-0.1164
Mean			-0.0074	-0.0089	-0.0261

DEMEANED, $\alpha = 0.35$

Window	Champion	Reconciled	Δ RMSE	Δ MAE	Δ High-vol
primary_spring_2016	2.0929	2.0890	-0.0040	-0.0011	-0.0106
christmas_2015	2.1547	2.1599	+0.0052	+0.0041	+0.0320
summer_2015	2.1074	2.0916	-0.0159	-0.0033	-0.0660
autumn_2015	2.1496	2.1289	-0.0208	-0.0034	-0.0961
Mean			-0.0088	-0.0009	-0.0352

Criterion	FULL	DEMEANED
H1 wins $\geq 3/4$	PASS (3/4)	PASS (3/4)
H2 mean Δ RMSE ≤ -0.005	PASS (-0.0074)	PASS (-0.0088)
H3 high-volume not worse	PASS (-0.0261)	PASS (-0.0352)
H4 mechanism $\geq 3/4$	FAIL (2/4)	FAIL (2/4)
H5 leakage checks	PASS	PASS
Decision	REJECTED	REJECTED

3.6 Why it fails, and why the failure is informative

The mechanism criterion asks whether the item-level model actually beats the champion's own bottom-up sum. It does so on only two of four windows — and the per-window outcome tracks that margin almost perfectly:

Window	Item model vs bottom-up	Δ RMSE (FULL)
christmas_2015	10.6% worse	+0.0158
primary_spring_2016	1.3% worse	-0.0017
summer_2015	0.3% better	-0.0187
autumn_2015	3.5% better	-0.0250

This is exactly what reconciliation theory predicts, and it is the reason to reject rather than a reason to hope. The physics is right; the input is not reliably available. The champion's bottom-up sum is **already** a strong item-level forecaster, and a dedicated item-level model beats it only sometimes.

This **converges with earlier project evidence from a completely different angle**. Experiment #76 measured a joint oracle of only -0.0055 for cross-store and cross-item **features** and rejected them before training. This branch attacked the same information channel through formal reconciliation with a separately trained aggregate model and hit the same wall. Two independent methods agreeing is far stronger than either alone: the bottom-level model already extracts what cross-store information this dataset contains, through global pooling and the `item_id` categorical.

4. External covariates

4.1 Availability at the real forecast origin (d_1941)

Checked against the raw files for d_1942–d_1969, the genuinely unknown window.

Covariate	Available for all 28 days	Needs forecasting?	Source
wday, month, year, is_weekend	28/28	No	calendar.csv
event_name_1/2, event_type_1/2	28/28	No	calendar.csv
snap_CA/TX/WI	28/28	No	calendar.csv
sell_price	30,490 / 30,490 series, all 5 future weeks	No	sell_prices.csv
promotion	0/28 — field does not exist	n/a	—

Events genuinely present in the forecast window: MemorialDay, NBAFinalsStart, NBAFinalsEnd, Ramadan starts, Father's day. Future price is used legitimately: the dataset publishes it, so the forecasting scenario genuinely assumes it is known at forecast time. No future actual-sales-derived quantity is used anywhere.

4.2 Residual structure — is anything left to explain?

A covariate whose residuals are already flat has nothing left to give.

Discount against the 52-week regular price (reference frozen at the origin — the retail-standard promotion proxy, and a genuinely different quantity from the week-on-week price changes already rejected in opt_02_v2_C_price):

Bin	n	Actual	Predicted	Residual	Share of sq. error
price ≥ regular	2,705	1.999	2.089	−0.090	0.49%
0–2% off	770,926	1.412	1.412	+0.000	88.94%
2–5% off	19,290	1.628	1.579	+0.049	2.48%
5–10% off	29,506	1.407	1.291	+0.115	2.52%
10–20% off	27,877	2.230	2.119	+0.111	5.13%
20–30% off	2,210	1.035	1.050	−0.015	0.17%
30%+ off	1,206	0.513	0.755	−0.242	0.26%

89% of rows sit in the "no meaningful discount" bin with a residual of +0.000. Events (+0.014 vs −0.025) and SNAP (+0.007 vs +0.010) are equally flat.

Oracle for a perfect discount correction, fitted on the evaluation window itself and therefore unreachable:

Correction	ΔRMSE
per discount bin	−0.0002
per weeks-at-current-price bin	−0.0012
per discount × price-age cell	−0.0019

Rejected without training. A promotion proxy cannot pay for itself when a perfect version of it is worth −0.0002.

One real but small effect is documented rather than fixed: items repriced within the last week are badly under-forecast (actual 1.706 vs predicted 0.711, residual +0.995) — but they are 0.46% of rows, so perfect treatment is worth −0.0012 overall.

5. Intermittent demand

Syntetos-Boylan classification from 728 days of pre-origin history (ADI cut 1.32, CV² cut 0.49):

Regime	Series	Rows	Zero %	Mean actual	RMSE	Bias	Share of sq. error
smooth	2,216	62,048	11.7	5.993	4.1661	−0.0077	28.82%
erratic	673	18,844	14.7	5.467	5.3957	−0.0981	14.68%
intermittent	23,247	650,916	60.9	0.831	1.2790	−0.0093	28.50%
lumpy	4,342	121,576	47.7	1.774	2.9326	+0.0133	27.98%
never sold	12	336	67.3	0.530	1.0428	−0.4199	0.01%

5.1 Croston, SBA and TSB — measured, not assumed

Each fitted on pre-origin history only ($\alpha = 0.1$), producing one rate per series across all 28 days, which is what these methods are.

Regime	Champion	Croston	SBA	TSB	Rolling mean 28
smooth	4.1661	4.5121	4.4982	4.5641	4.6191
erratic	5.3957	5.5322	5.5126	5.5913	5.7174
intermittent	1.2790	1.3781	1.3741	1.3417	1.3625
lumpy	2.9326	3.1407	3.1343	3.0217	3.0706
ALL	2.0920	2.2380	2.2317	2.2084	2.2430

There is no regime in which any classical intermittent method beats the champion. Not one.

5.2 Is Tweedie adequate?

A wrong likelihood shows up as regime-dependent bias. Bias is flat: −0.13% of mean demand on smooth, −1.79% erratic, −1.12% intermittent, +0.75% lumpy. The only outlier is the 12 "never sold" series (0.01% of squared error).

5.3 Oracles for specialisation

Upper bound (fitted on the evaluation window)	Δ RMSE
per-regime multiplicative rescale	−0.0008
per-regime blend with Croston	−0.0048
per-regime blend with SBA	−0.0046
per-regime blend with TSB	−0.0003
per-regime blend with rolling mean 28	+0.0000

Combined with the existing evidence — hurdle tested twice (2.1267, 2.1241, both worse), intermittency-quintile segmentation oracle −0.0025 — the single Tweedie head is confirmed adequate. **Requirement C is satisfied by Tweedie, and that is now a measured claim rather than an assumption.**

6. Why a candidate that improved three metrics was still rejected

Experiment #81's DEMEANED variant improved mean RMSE (−0.0088), mean MAE (−0.0009) and mean high-volume RMSE (−0.0352) at once — something almost nothing in this project's 79 prior experiments achieved. It was rejected anyway. The reasoning matters more than the number.

- **The pre-registered mechanism criterion failed.** H4 was fixed before the run precisely to prevent accepting a gain whose cause cannot be verified. Waiving it after seeing a favourable mean is the exact failure mode the research brief forbids.
- **The gain is not attributable.** It tracks whether the item-level model happens to beat the bottom-up sum that month — knowable only after the fact. The expected gain at the real forecast origin is therefore not −0.0088; it is a draw from a distribution containing +0.0158.
- **It harms the operationally most important window.** Christmas is where retail forecasting matters most, and FULL costs +0.0158 RMSE and +0.0820 high-volume RMSE there.
- **The incumbent was held to a higher bar.** The shipped blend was accepted on 4/4 windows, 3/3 seeds, with a negative control isolating the cause. Promoting a 3/4 result with a failed mechanism test would lower the project's evidential standard, not raise its accuracy.

What would overturn this. An item-level model that beats the bottom-up sum on 3+ of 4 windows, or a leakage-safe pre-origin signal that reliably predicts which regime the next 28 days fall into. Experiment #82 tested the second of these directly, and it failed.

6.1 Experiment #82 — the deployable per-origin α , and why it does not rescue the method

At any forecast origin $*T*$ the 28 days ending at $*T*$ are fully observed, so α can be chosen on $(T-28, T]$ and applied to $(T, T+28]$. At the real forecast origin d_1941 the selection window is d_1914–d_1941 — exactly the data a practitioner standing on 2016-05-22 would hold. This is a complete, leakage-safe, deployable rule, and it was pre-registered before Experiment #81 produced any result.

Window	α selected on	FULL α	FULL Δ RMSE	DEMEANED α	DEMEANED Δ RMSE
primary_spring_2016	d_1885	0.55	−0.0017	0.35	−0.0040
christmas_2015	d_1750	0.60	+0.0198	0.50	+0.0145
summer_2015	d_1601	0.80	−0.0151	0.65	−0.0187
autumn_2015	d_1679	0.80	−0.0275	0.75	−0.0291
Mean			−0.0061		−0.0093

Criterion	FULL	DEMEANED
K1 wins $\geq 3/4$	PASS (3/4)	PASS (3/4)
K2 mean Δ RMSE ≤ -0.005	PASS (−0.0061)	PASS (−0.0093)
K3 high-volume not worse	PASS (−0.0229)	PASS (−0.0406)
K4 α spread ≤ 0.40	PASS (0.25)	PASS (0.40)
K5 mechanism $\geq 3/4$	FAIL (2/4)	FAIL (2/4)
Decision	REJECTED	REJECTED

The decisive observation is not the mean — it is which way α moved. Going into christmas_2015, the one window where the item-level model collapses (10.98 versus 9.92 for the bottom-up sum), the preceding window told the rule to be **more**

aggressive: α rose from 0.55 to 0.60 (FULL) and from 0.35 to 0.50 (DEMEANED). The loss consequently grew from +0.0158 to +0.0198 and from +0.0052 to +0.0145.

A working detector would have shrunk α toward zero there. Instead the adaptive rule amplified the harm, because **the item model's advantage does not persist from one 28-day window to the next**. The selected α also swings across half the grid (0.55 $\rightarrow$ 0.60 $\rightarrow$ 0.80 $\rightarrow$ 0.80) without tracking the quantity it needs to.

Against the fixed- α protocol the adaptive one is a wash — FULL is slightly worse (−0.0061 vs −0.0074), DEMEANED slightly better (−0.0093 vs −0.0088) — so the extra machinery buys nothing.

This is the strongest form the negative result could take. The rejection is not "we could not tune it well enough"; it is that **the quantity you would have to tune on is not predictable from the information available at forecast time**.

7. Use Case 11 final compliance matrix

Requirement	Status	Evidence
Hierarchical aggregation	✓	Bottom-up coherent by construction; accuracy measured at all 12 M5 levels (uc11_hierarchy_levels.csv); reconciliation investigated formally at 3 levels with exact oracle bounds; 4-window validation (#81); rejected on evidence, not omitted
External price covariates	✓	4 price features; 30,490/30,490 series priced for all 28 future days; residuals flat; perfect-correction oracle -0.0002
Promotion covariates	~	No promotion field exists in M5. Represented by price-relative proxies; discount-vs-52-week-regular audited, oracle -0.0002; rejected on measurement
Holiday / calendar covariates	✓	4 event fields + weekday/month/year/weekend + state-matched SNAP; 28/28 days available at the real origin; residuals flat
Intermittent demand	✓	Tweedie(1.1) by power sweep; hurdle rejected twice; Croston/SBA/TSB now measured and beaten in every regime; bias flat across regimes; specialisation oracle -0.0008
28-day-ahead forecasting	✓	Fixed-origin direct 28-day; 46 foundation checks; 9999-corruption test on both the champion's builder and the 30 new aggregate features; recursive rollout structurally verified per window
Per store-item forecasting	✓	853,720 predictions (30,490 × 28), none excluded or weighted
Leakage-safe	✓	Structural + empirical corruption tests; all training origins satisfy $\text{origin} + \text{horizon} \leq \text{validation_origin}$; α and every constant selected on pre-origin windows only
Validated accuracy	✓	4 disjoint windows, multiple seeds, paired comparisons on identical rows; champion reproduced to 4 decimals on all 4 windows in this branch

One requirement remains PARTIAL, and it cannot be made otherwise: M5 has no promotion field. Marking it satisfied would require inventing a covariate the data does not contain.

8. What must not change

- **The champion configuration.** $0.60 \times \text{direct}(38f) + 0.40 \times \text{recursive}(32f)$, Tweedie(1.1), 400 rounds, seed 42. Reproduced exactly four times in this branch.
- **The blend weight $w = 0.60$,** selected on an inner window in #77.
- **The Tweedie objective at the bottom level.** Now confirmed against Croston/SBA/TSB in every demand regime, not just in aggregate.
- **The fixed-origin direct-28-day design and its corruption tests.**
- **The validation protocol** — four disjoint windows, pre-registered criteria, selection on pre-origin windows only.
- **The raw data.** All five MD5s re-verified unchanged during this branch.

Directions now closed on measured evidence

Direction	Evidence	Verdict
Reconciliation above item level	true-aggregate oracle ≤ -0.0221	dead
Item-level reconciliation, fixed α	3/4 windows but mechanism 2/4	rejected
Item-level reconciliation, adaptive α	3/4 windows, mechanism 2/4, α moves the wrong way on the failing window	rejected
Cross-store / cross-item features	joint oracle -0.0055 (#76)	dead
Promotion / discount covariates	perfect-correction oracle -0.0002	dead
Croston / SBA / TSB	lose in every regime	dead
Regime segmentation	oracle -0.0008	dead
Hurdle models	2.1267 / 2.1241, both worse	dead
Global or per-series recalibration	-0.0014 global; #69 failed	dead

9. Limitations

- **The promotion covariate does not exist** in this dataset and no feature here recovers it.
 - **Stockouts remain unobservable.** A zero can mean "nobody wanted it" or "it was not on the shelf", and nothing distinguishes them.
 - **Item-level reconciliation is rejected, not disproven.** It helps materially on two of four windows, and both protocols improved mean RMSE, MAE and high-volume RMSE together. What defeats it is that its benefit is not predictable at forecast time — Experiment #82 showed the obvious pre-origin signal moves α the wrong way precisely when it matters. A genuinely predictive regime signal would revive the direction.
 - α **selected on a single window proved unreliable**, and the adaptive replacement was no better. Both are documented rather than hidden.
 - **Point forecasts only.** No predictive intervals, which inventory decisions want.
 - **Documentation drift.** README.md, PROJECT_INDEX.md, models/champion/README.md and predictions/final_forecast/README.md still describe the 32-feature single model at RMSE 2.1210 as the champion, and the final-forecast README quotes 2.1210 as the estimate for the delivered window. The delivered file is final_forecast_28day_v3_diversity_blend.csv from the 0.60/0.40 blend at 2.0929. These files were left untouched by this branch; correcting them is a separate, deliberate decision.
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10. Artefacts produced

New code (nothing existing was modified):

Path	Purpose
pipeline/aggregate_level.py	aggregate-level panel + leakage-safe feature builder
pipeline/champion_blend.py	faithful reproduction of the shipped champion, with caching
scripts/07_usecase11/50_hierarchy_headroom.py	12-level oracle diagnostic
scripts/07_usecase11/51_reconciliation_threshold.py	break-even curve
scripts/07_usecase11/52_aggregate_model_target.py	required aggregate accuracy
scripts/07_usecase11/53_exp80_item_level_probe.py	inner-window go/no-go
scripts/07_usecase11/54_exp80b_level_sweep.py	level sweep + negative controls
scripts/07_usecase11/56_exp80c_orthogonality.py	level vs cross-store decomposition
scripts/07_usecase11/57_exp81_four_window_validation.py	four-window validation
scripts/07_usecase11/59_exp82_adaptive_alpha.py	per-origin α protocol
scripts/07_usecase11/55_covariate_audit.py	requirement-6 audit
scripts/07_usecase11/58_intermittency_audit.py	requirement-7 audit
scripts/07_usecase11/60_record_experiments.py	registry bookkeeping

Artefacts are experiments/artifacts/uc11_*; cached champion reproductions are predictions/uc11_cache/. No file under data/raw/, models/champion/, predictions/final_forecast/ or experiments/registry/ was modified.